北京交通大學 BEIJING JIAOTONG UNIVERSITY

信号与系统

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Time domain analysis of continuous LTI system

- Method based on solving differential equations for linear constant coefficients
- Methods based on zero input responses versus zero state responses
- Time domain representation of continuous LTI systems—impulse response

Method based on solving differential equations for linear constant coefficients

Continuous LTI systems can be described by differential equations of linear constant coefficients

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)$$
$$=b_mx^{(m)}(t)+b_{m-1}x^{(m-1)}(t)+\cdots+b_1x'(t)+b_0x(t)$$

The full solution of a differential equation consists of a homogeneous solution of $y_h(t)$ and a special solution of $y_p(t)$.

$$y(t)=y_h(t)+y_p(t)$$

- ✓ The form of the homogeneous solution $y_h(t)$ is determined by the eigengene root corresponding to the differential equation
- ✓ The form of the special solution $y_p(t)$ is determined by the form of the excitation signal on the right side of the equation

The full solution of the differential equation is the output response of the system.

Method based on solving differential equations for linear constant coefficients

[Example] Differential equations that describe a continuous LTI system are known

$$y''(t)+6y'(t)+8y(t)=x(t),\ t>0$$

Initial $y(0)=1$, $y'(0)=2$, input signal $x(t)=e^{-t}u(t)$, solve the solution $y(t)$

(1) Determine the homogeneous equation $y''(t)+6y'(t)+8y(t)=0$ of the homogeneous equation $y_h(t)$

The characteristic equation is $s^2+6s+8=0$

The feature root is $s_1=-2,\ \ s_2=-4$

Solution $y_h(t)$ $\quad y_h(t)=Ae^{-2t}+Be^{-4t} \qquad t>0$

Method based on solving differential equations for linear constant coefficients

[Example] Differential equations that describe a continuous LTI system are known

$$y''(t)+6y'(t)+8y(t)=x(t),\ t>0$$

Initial $y(0)=1$, $y'(0)=2$, input signal $x(t)=e^{-t}u(t)$, solve the solution $y(t)$

(2) Solve the special solution of the differential equation y''(t)+6y'(t)+8y(t) = x(t)

In the form of input x(t), the special solution of the equation is set $y_p(t) = Ce^{-t}$ $t>0$

Bring the special solution into the original differential equation to find the coefficients to be determined $C=1/3$。

Method based on solving differential equations for linear constant coefficients

[Example] Differential equations that describe a continuous LTI system are known

$$y''(t)+6y'(t)+8y(t)=x(t),\ t>0$$

Initial $y(0)=1$, $y'(0)=2$, input signal $x(t)=e^{-t}u(t)$, solve the solution $y(t)$

(3) Find the full solution of the differential equation

$$y(t)=y_h(t)+y_p(t)=Ae^{-2t}+Be^{-4t}+e^{-t}/3$$

$$y(0)=A+B+1/3=1$$

$$y'(0)=-2A-4B-1/3=2$$

$\Rightarrow$ $A=5/2$, $B=-11/6$

$$y(t)=\frac{5}{2}e^{-2t}-\frac{11}{6}e^{-4t}+\frac{1}{3}e^{-t},\ t>0$$

Method based on solving differential equations for linear constant coefficients

(1) If the initial conditions do not change, the input signal changes to $x(t) = e^{-2t}u(t)$, the system is fully responsive $y(t) = ?$

Solution: (1) Determines the form of the homogeneous equation

y''(t)+6y'(t)+8y(t)=0 of the homogeneous solution $y_h(t)$.

(2) Find the special solution $y_p(t)$ of the equation y'(t)+6y'(t)+8y(t) = x(t)

(3) Find the full solution of the equation

Method based on solving differential equations for linear constant coefficients

(2) If the input signal does not change, the initial condition changes to y(0) = -1, y '(0) = 1, the system is fully responsive $y(t)$ = ?

Solution: (1) Determines the form of the homogeneous equation y''(t)+6y'(t)+8y(t)=0 of the homogeneous solution $y_h(t)$.

(2) Find the special solution $y_p(t)$ of the equation y'(t)+6y'(t)+8y(t) = x(t)

(3) Find the full solution of the equation

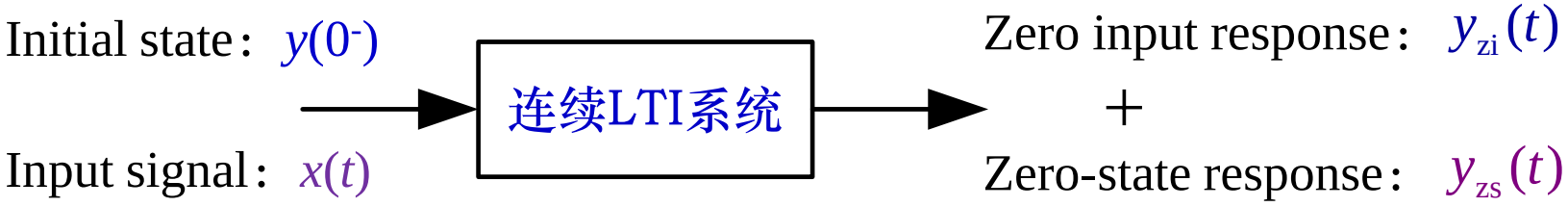
Method based on solving differential equations for linear constant coefficients

Inadequacies of methods based on solving differential equations of constant coefficients

- If the input signal changes, it must be fully resolved.
- If the initial conditions change, they must all be resolved.
- If the differential equation excitation term is more complex, it is difficult to deal with.
- The physical concept of continuous LTI system response is not represented.

Methods based on zero input responses versus zero state responses

According to the linear nature of the LTI system, the system response is considered by the system. The superposition of the response resulting from the initial state and the input signal acting separately on the system.



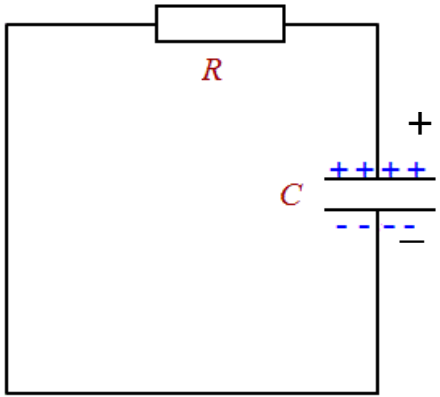
Fully responsive $y(t)$ = Zero input response $y_{zi}(t)$ + Zero-state response $y_{zs}(t)$

Methods based on zero input responses versus zero state responses

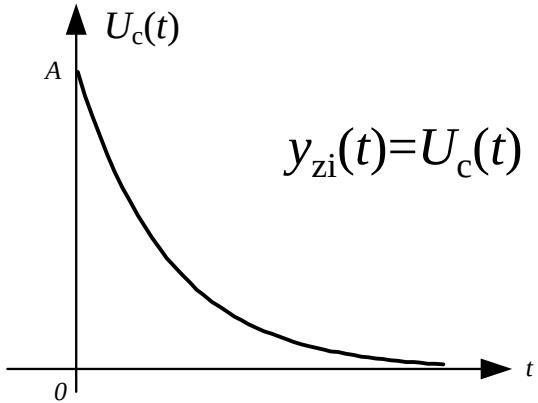
Zero input response

The response produced by the input signal being zero and acting on the system alone by acting solely on the system only by the initial state of the system is called the zero input response, denoted as $y_{zi}(t)$.

The system has no external inputs, only the initial energy storage of the capacitor.



Capacitive discharge system



Schematic of capacitance voltage

Methods based on zero input responses versus zero state responses

Zero input response

The linear constant coefficient differential equation that describes a continuous LTI system is

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)$$
$$=b_mx^{(m)}(t)+b_{m-1}x^{(m-1)}(t)+\cdots+b_1x'(t)+b_0x(t)$$

When input $x(t)=0$

$$y^{(n)}(t)+a_{n-1}y^{(n-1)}(t)+\cdots+a_1y'(t)+a_0y(t)=0$$

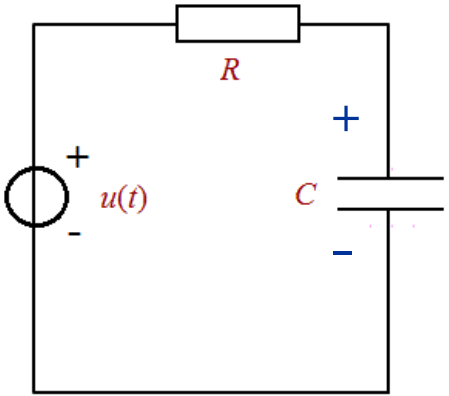
The above equation is homogeneous, so the zero input response has the form of a homogeneous solution

Methods based on zero input responses versus zero state responses

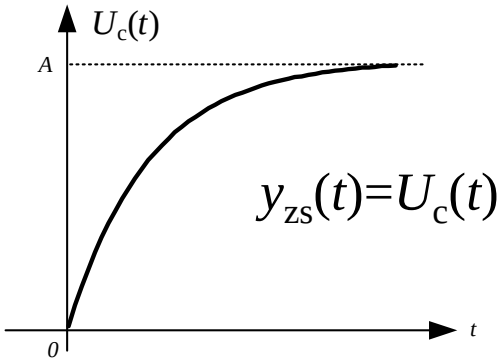
Zero-state response

The initial state is zero, and the response generated only by the external signal x(t) of the system is called the zero-state response of the system, denoted by $y_{zs}(t)$

The capacitor has an initial energy reserve of zero and is produced only by an external input.



Capacitive charging system



Schematic of capacitance voltage

Methods based on zero input responses versus zero state responses

Zero-state response analysis method

(1) Represents the input signal as a linear combination of unit impulse signals

(2) Find out the zero-state response generated by the unit impulse signal acting on the system - Impulse response h(t)

(3) Using the characteristics of linear non-time-varying systems, the input signal x(t) can be found. The zero-state response of the system under excitation is $y_{zs}(t)$.

Methods based on zero input responses versus zero state responses

Zero-state response

Non-time-varying characteristics:

Uniformity:

Overlay:

$\Delta\tau \to 0$:

$$\delta(t) \longrightarrow \boxed{\text{连续LTI系统}} \longrightarrow h(t)$$

System impulse response

$$\delta(t-k\Delta\tau) \longrightarrow \boxed{\text{连续LTI系统}} \longrightarrow h(t-k\Delta\tau)$$

$$x(k\Delta\tau)\Delta\tau\cdot\delta(t-k\Delta\tau) \longrightarrow \boxed{\text{连续LTI系统}} \longrightarrow x(k\Delta\tau)\Delta\tau\cdot h(t-k\Delta\tau)$$

$$\sum_{k=-\infty}^{+\infty} x(k\Delta\tau)\Delta\tau\cdot\delta(t-k\Delta\tau) \longrightarrow \boxed{\text{连续LTI系统}} \longrightarrow \sum_{k=-\infty}^{+\infty} x(k\Delta\tau)\Delta\tau\cdot h(t-k\Delta\tau)$$

$$\int_{-\infty}^{+\infty} x(\tau)\delta(t-\tau)\mathrm{d}\tau \longrightarrow \boxed{\text{连续LTI系统}} \longrightarrow \int_{-\infty}^{+\infty} x(\tau)h(t-\tau)\mathrm{d}\tau$$

$x(t)$

$$y_{zs}(t) = x(t) * h(t)$$

Methods based on zero input responses versus zero state responses

System response $y(t)$ = Zero input response $y_{zi}(t)$ + Zero-state response $y_{zs}(t)$

Solve homogeneous differential equations

$$y_{zs}(t) = x(t) * h(t)$$

※ Zero input response solution

※ Solving the impulse response

※ Zero-state response solution

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!